pfi

Python for Fixed Income

COURSE 3 · MARKET DATA AND TIME SERIES

JSON and the FRED API

WEEK 1 · DAY 2

Created by **Jeff Lenamon**

This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

From one cell to three functions

`fred_params`

Builds the request

no network

`parse_fred_observations`

Reads the answer

no network

`fred_series`

Sends it, then parses

the one network function

Two of the three need no network, so they are tested on a sample file, with no key, in a fraction of a second.

Your key never shows: not in a printed URL, an error message, or a traceback.

Week 1: getting the data in

Day 1	APIs, keys, and secrets
Day 2	**JSON and the FRED API**
Day 3	Market calendars, cleaning
Day 4	Resampling, the as-of value
Day 5	Case study: the dataset

JSON in one slide

```
{
  "series_id": "DGS10",
  "complete": true,
  "notes": null,
  "observations": [
    {"date": "2026-09-30", "value": "5.29"}
  ]
}
```

Two containers: an object { } of name and value pairs, and a list [ ] of values in order. Plain text, so any program can read it.

JSON	Python
object { }	dict
list [ ]	list
"text"	str
5 or 5.29	int or float
true, false	True, False
null	None

FRED sends every value as text: "5.29", not 5.29.

FRED's answer

FRED's answer: metadata beside a list of dates and text values

The answer: one JSON object

realtime_start:	2026-10-02
realtime_end:	2026-10-02
observation_start:	2026-08-03
observation_end:	2026-09-30
units:	lin
output_type:	1
file_type:	json
order_by:	observation_date
sort_order:	asc
count:	43
offset:	0
limit:	100000

observations: a list of 43 small objects

```
[ 0] {"date": "2026-08-03", "value": "4.70"}
[ 1] {"date": "2026-08-04", "value": "4.63"}
   ...
[24] {"date": "2026-09-04", "value": "4.78"}
[25] {"date": "2026-09-07", "value": "."}
[26] {"date": "2026-09-08", "value": "4.80"}
   ...
[42] {"date": "2026-09-30", "value": "5.29"}
```

The orange row: a weekday with no value

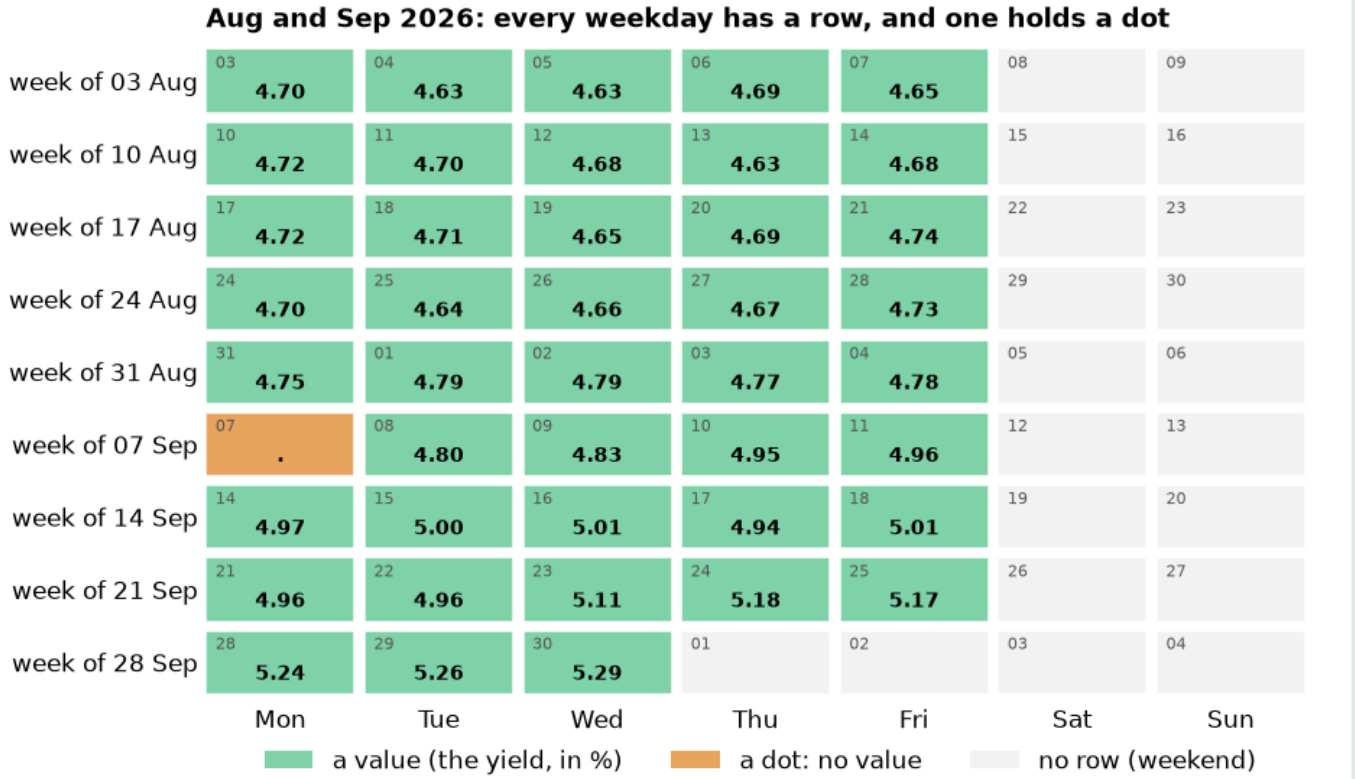
The sample file

FRED's layout, Treasury's values, built for the course. Nothing in it was downloaded from FRED. No key, no network.

FRED Help: "Before the popularization of the World Wide Web, the data were provided in list form on a dial-in, electronic bulletin board system."

FRED Help, What is FRED? (read 2026-10-04)

A row is not a number



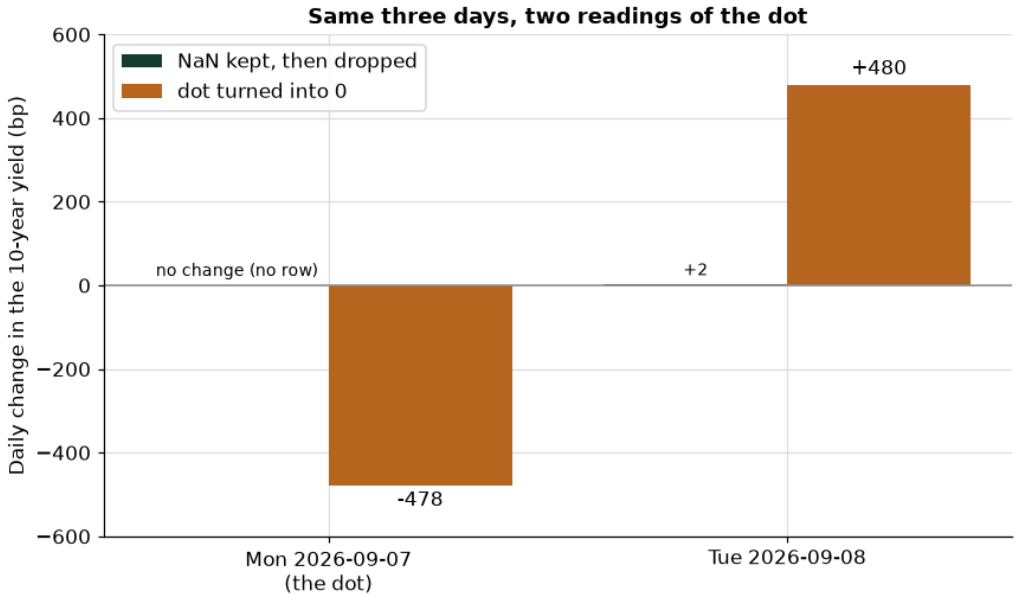
Rows and numbers

43
rows

42
numbers

A weekday with no value still has a row. Weekends have none. The orange square is Labor Day.

A dot is missing, never zero



A dot becomes NaN, never a number. NaN rows are dropped, in a step you can see, before any change or window.

Friday 4 Sep: 4.78. Tuesday 8 Sep: 4.80. Kept as NaN, the change is 2 bp from Friday. As 0: -478 bp, then +480 bp.

4.8367%

mean of 42 days, NaN left out

4.7242%

mean of 43 days, dot as 0

11.2 bp

moved by one zero

fred_params and parse_fred_observations

fred_params: pure, sends nothing

```
md.fred_params("DGS10", FAKE_KEY,
               start="2026-08-03")
# holds the key: never printed
```

parse_fred_observations

```
float("nan") if obs["value"] == "."
else float(obs["value"])
# never 0
```

43

rows, one NaN, float64

42

after dropna, a visible step

0.0

largest gap to the Treasury file

7

tests passed

No observations in the answer (one of FRED's error answers): ValueError, naming the keys it found.

fred_series: errors that hide the key

response.raise_for_status() says, with the fake key:

```
400 Client Error: Bad Request for url: ...
  ...?series_id=DGS10&api_key=fakefakefakefakefakefakefakefake...
```

1. Never raise_for_status
2. Its own error: status and FRED's message
3. Connection error: the class name only
4. Raised from None: no chained traceback
5. Empty answer, status 200: stop

Tests with monkeypatch and a stand-in answer: no fake key and no api_key= anywhere in the printed error. 9 passed.

Live: the 10-year and the 2-year from 2015

```
for live_id, live_column in [("DGS10", "10 Yr"),
                            ("DGS2", "2 Yr")]:
    live_series = md.fred_series(live_id, fred_key, ...)
```

A live cell, two ways

No key (the saved notebook):

```
Live cell skipped: no FRED key was found.
```

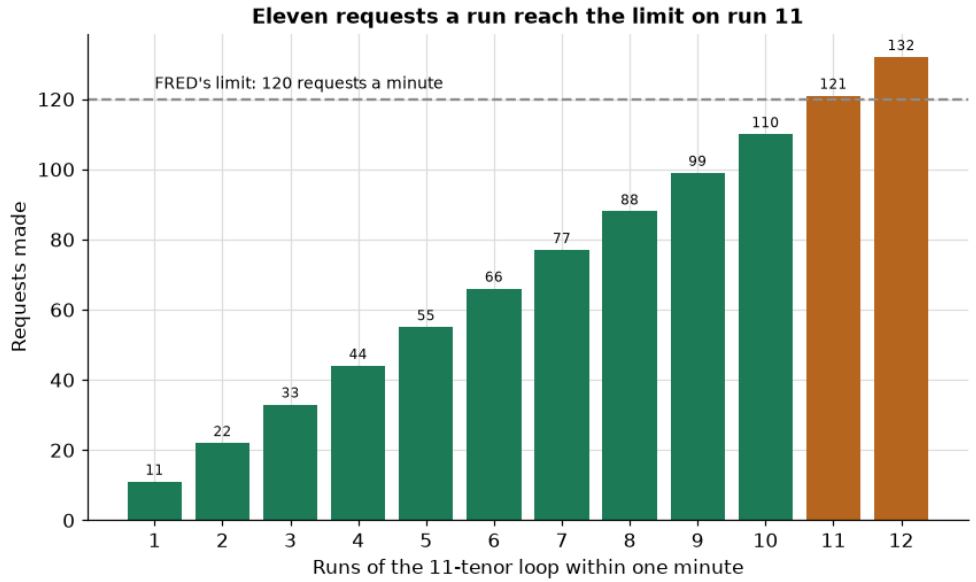
With your key:

one line per series, largest gap 0

1. Equal on every common date (1e-9 percent)
2. Every dot on a date the file has no row for
3. No FRED number on a date the file lacks

Checked 2026-10-04: the Treasury file ends 2026-10-02, FRED's last DGS10 was 2026-10-01. Only common dates are compared.

The rate limit and the notice



120

requests a minute; then HTTP 429

3

requests in today's live cells

The whole curve, one tenor at a time, is 11 requests a run. A loop that reruns it reaches the limit quickly.

This product uses the FRED® API but is not endorsed or certified by the Federal Reserve Bank of St. Louis.

FRED API Terms of Use and FRED API errors page (read 2026-10-04).

Excel side by side: Power Query reads the JSON

```
Source = Json.Document(File.Contents("...json"))
Rows = Table.FromRecords(Source[observations])
NoDots = Table.SelectRows(Rows, each [value] <> ".")
```

Query	Excel returns
Read the JSON	43 rows, value as text
Type to number	43 rows, the dot an empty cell
SelectRows first	42 rows, all numbers

A change across the gap

`=(D28-D27)*100 on 2026-09-08`

480 bp

The empty cell counts as 0: the same wrong change as Python's dot made 0.

Tested in Excel 16.0 build 20527, 2026-11-02, on the course's sample file.

Excel side by side: VALUE and IFERROR

Excel side by side (tested in Excel)

`=VALUE(".")`	#VALUE! like float(".") stopping
`=IFERROR(VALUE(B2),NA())`	#N/A like errors="coerce" giving NaN
`=B3-B2`	#VALUE! a number minus the text
`=AVERAGE(B2:B4)`	4.23 text is skipped, COUNT 1

B2 the text ".", B3 the number 4.23, B4 the text '4.25.

Leave B2 empty and =B3-B2 returns 4.23: the empty cell counts as 0. The spreadsheet version of today's bug.

Save the day: read before you commit

git diff --cached

What the commit is about to record: the staged change, line by line. Today every line starts with +.

```
3 files changed, 452 insertions(+)
```

git grep -n -E "[a-z0-9]{32}"

Every tracked or staged file, searched for the shape of a FRED key. Your ignored .env is never read.

```
test_marketdata.py:12:FAKE_KEY = "fakefake...fake"
```

One hit, the course's fake key: expected. A real key there: take it out if only staged; rotate it if committed.

The AI check

The docstring is the prompt. The code that comes back runs, and has one data bug.

1

Docstring

Units, conventions, and what it must never do: what you asked for.

2

Code back

Written for the course in the style of an AI answer. Read it.

3

Test first

Assert what the docstring promises. Run it: it fails.

4

Fix

Find the rule it broke, fix it, run the test again.

Today: parse_yields turns "." into 0.0 "so the column stays numeric": 2026-09-07 reads as a 0 percent yield.

Code that runs is not the same as code that is right.

What you can do now

- JSON: objects and lists; json.loads gives dictionaries and lists
- FRED's answer: metadata, count, and observations whose values are text
- A dot is NaN, never 0, dropped in a visible step
- fred_params and parse_fred_observations, tested with no network
- fred_series never calls raise_for_status; read the diff, scan for keys

Week 1, Day 3 Market calendars and cleaning: weekdays with no row, holiday calendars, and joins that invent nothing